

MAT1100

Aaron Avram

September 8th 2026

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1 Group Theory

1.1 Basic Definitions and Results

Definition 1. A Group is a set with a binary operation $\star : G \times G \rightarrow G$ satisfying

1. Associativity: for all $a, b, c \in G$, $(a \star b) \star c = a \star (b \star c)$
2. Existence of an identity: $\exists e \in G$ such that for all $a \in G$, $a = a \star e = e \star a$.
3. Existence of inverses: for all $a \in G$ there exists $b \in G$ such that $ab = ba = e$.

Example 1. Some examples

1. $(\mathbb{Z}, +)$
2. $GL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) \neq 0\}$
3. $SL_n(\mathbb{K}) = \{A \in M_n(\mathbb{K}) \mid \det(A) = 1\}$

Example 2. D_8 the group of symmetries of the unit square S .

$$D_8 = \{T \in GL_2(\mathbb{R}) : T(S) = S \wedge \|T(x) - T(y)\| = \|x - y\|, \forall x, y \in \mathbb{R}^2\}$$

it is not hard to check that this forms a group under function composition.

Definition 2. Subgroups $H \subset G$ is a subgroup and denoted $H \leq G$ if it is a group under the operation of G .

Equivalently:

1. $e \in H$
2. $xy \in H$ for all $x, y \in H$
3. $x^{-1} \in H$ for all $x \in H$.

Remark 1. There is a subgroup criterion that is sometimes useful: $H \subseteq G$ is a subgroup if and only if $H \neq \emptyset$ and for all $x, y \in H$, $xy^{-1} \in H$

Example 3. Some examples

1. $SL_n(\mathbb{K}) \leq GL_n(\mathbb{K})$
2. $(\mathbb{Z}, +) \leq (\mathbb{Q}, +)$
3. $D_8 \leq GL_2(\mathbb{R})$.

Definition 3. For $g \in G$ a group, the cyclic subgroup generated by g is

$$\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$$

Example 4. $S_3 := \{\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\} \text{ a bijection}\}$

$$\langle (1, 2) \rangle = \{e, (1, 2)\}$$

and

$$\langle (1, 2, 3) \rangle = \{e, (1, 2, 3), (1, 3, 2)\}$$

Definition 4. The order of a group G is its cardinality and denoted $|G|$. For $g \in G$, $|g| = \min\{n \in \mathbb{N} : g^n = e\}$ called the order of g .

Proposition 1. For $g \in G$ we have $|\langle g \rangle| = |g|$.

Proof. Set $n = |g|$. By the minimality of the order $e, g, \dots, g^{n-1}$ are all distinct. Now it suffices to show that $\langle g \rangle = \{e, g, \dots, g^{n-1}\}$ the reverse containment is clear. For the forward containment, let $g^a \in \langle g \rangle$. $a \in \mathbb{Z}$, so there exists $k \in \mathbb{Z}$ and $r \in 0, \dots, n-1$ such that

$$a = kn + r$$

then

$$g^a = g^{kn+r} = g^{kn}g^r = g^r$$

hence g^a is represented by an element in the right hand set which proves the proposition. $\square$

Definition 5. A group homomorphism is a map $\varphi : G \rightarrow G'$ such that $\varphi(gh) = \varphi(g)\varphi(h)$ for all $g, h \in G$.

1. The kernel of φ : $\ker \varphi = \{g \in G | \varphi(g) = e_{G'}\}$
2. The image of φ : $\text{Im } \varphi = \{\varphi(g) \mid g \in G\}$

Proposition 2. $\ker \varphi \leq G$ and $\text{Im } \varphi \leq H$.

Example 5. $\varphi = \det : GL_n(\mathbb{K}) \rightarrow \mathbb{K}^\times$ with $\ker \varphi = SL_n(\mathbb{K})$.

Example 6. Now we compute D_8 explicitly. Label the vertices of the square 1, 2, 3, 4 starting from the bottom left and moving counter clockwise.

A symmetry of the square must be a permutation as the vertices are exactly the points of the square which have another point a distance of the diagonal away and isometries must preserve this.

Additionally isometries must preserve adjacency

Hence given an element $T \in D_8$, $T(1)$ has 4 possibilities, then given $T(1)$, $T(2)$ has 2 and then by adjacency the isometry is now determined.

Therefore $|D_8| \leq 8$.

Now it is clear that two isometries are:

1. s: reflection about the diagonal 1-3.
2. r: counter-clockwise rotation by $\frac{\pi}{2}$

and from here it is easy to show that

$$D_8 = \{e, s, r, r^2, r^3, rs, r^2s, r^3s\}$$

and from here we recover another group as a subgroup

$$\{e, s, r^2, r^2s\} = K_4$$

the Klein 4-group.

1.2 Group Actions

Definition 6. The action of a group G on a set $X \neq \emptyset$ is a map

$$G \times X \rightarrow X$$

given by

$$(g, x) \mapsto g \cdot x$$

such that

$$e \cdot x = x \quad \forall x \in X$$

and

$$(gh) \cdot x = g \cdot (h \cdot x) \quad \forall x \in X$$

Example 7.

(1) $D_8 \curvearrowright V$ where $V = \{1, 2, 3, 4\}$ are the vertices of the unit square. This action is given by:

$$(g, i) \mapsto g \cdot i = \sigma_g(i)$$

where σ_g is the permutation of the vertices associated with g .

(2) $D_8 \curvearrowright \Delta = \{d_1, d_2\}$

$$g \cdot \{i, j\} = \{g(i), g(j)\}$$

by adjacency.

Theorem 2. Let $G \curvearrowright X$ be an action of a group G on a set X . Then for each $g \in G$ the map

$$\rho(g) : X \rightarrow X$$

given by

$$\rho(g)(x) = g \cdot x$$

is a bijection with inverse $\rho(g^{-1})$ and the map

$$\rho : G \rightarrow \text{Sym}(X)$$

given by:

$$\rho : g \mapsto \rho(g)$$

is a group homomorphism.

Remark 3. The converse is also true, a homomorphism $\rho : G \rightarrow \text{Sym}(X)$ induces an action.

Definition 7. For an action $G \curvearrowright X$ the orbit of $x \in X$ is $\mathcal{O}_x = \{g \cdot x : g \in G\}$. An action is transitive if it has only one orbit.

Proposition 3. The relation on X given by $a \sim b \iff \exists g \in G \text{ s.t. } g \cdot a = b$ is an equivalence relation, and its equivalence classes are the distinct orbits of X .

Proposition 4. *A subgroup $H \leq G$ induces an action $H \curvearrowright G$ given by*

$$h \cdot g = gh^{-1}$$

and its orbits

$$\mathcal{O}_g = \{h \cdot g : h \in H\} = \{gh^{-1} : h \in H\} = gH$$

is called the left coset of g in H . It is clear $|gH| = |H|$.

Definition 8. Write G/H to be the set of left cosets, then

$$[G : H] = \#(G/H)$$

Theorem 4. *If $H \leq G$ we have $|G| = |H|[G : H]$. In particular the order of a subgroup divides the order of G*

Corollary 5. *The order of an element divides the order of the group.*